

Dijkstra's Algorithm:-

1.

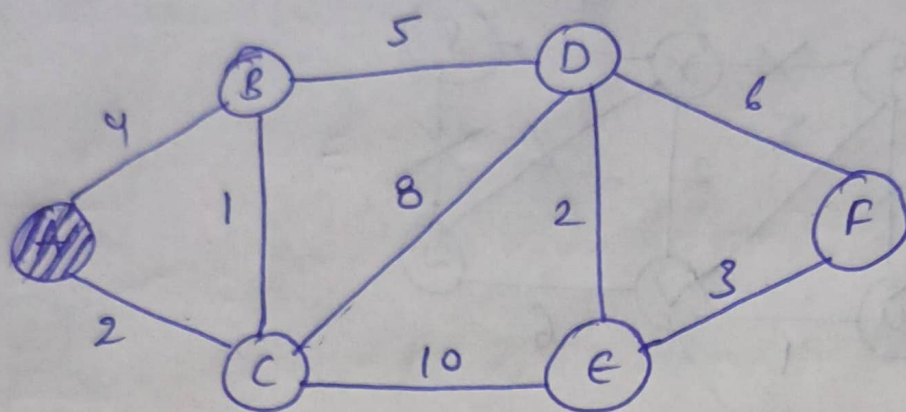
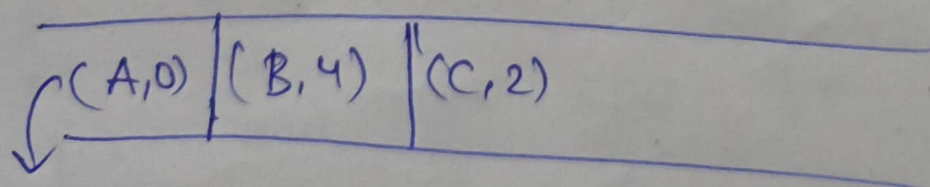


Table:-

Node	cost	previous
A	0	
B	4 3	A C
C	2	A
D	8	B
E	10	D
F	14	D

priority queue:-



i, at A :- $d(A) = 0$

$A \rightarrow B$: $0 + 4 < \infty$ ✓ , cost = 4

$A \rightarrow C$: $0 + 2 < \infty$ ✓ , cost = 2

(B, 4)	(C, 2)
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ii, at C :- $d(C) = 2$

$C \rightarrow A$: $2 + 2 < \infty$, cost not updated

$C \rightarrow B$: $2 + 1 < 4$, cost = 3

$C \rightarrow D$: $2 + 8 < \infty$, not updated

$C \rightarrow E$: $2 + 10 < \infty$, cost = 12

(B, 3)	(E, 12)
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iii, at B :- $d(B) = 3$

$B \rightarrow A$: $3 + 4 < \infty$, don't update

$B \rightarrow C$: $3 + 1 < 2$, don't update

$B \rightarrow D$: $3 + 5 < \infty$, cost = 8

(E, 12)	(D, 8)
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iv, at D :- $d(D) = 8$

$D \rightarrow B$: $8 + 5 < 3$, don't update

$D \rightarrow C$: $8 + 8 < 2$, don't update

$D \rightarrow E$: $8 + 2 < 12$, cost = 10

$D \rightarrow F$: $8 + 6 < \infty$, cost = 14

(E, 10) | (F, 14)

✓) at E :- $d(E) = 10$

$E \rightarrow C : 10 + 10 < 2$, don't update

$E \rightarrow D : 10 + 2 < 8$, don't update

$E \rightarrow F : 10 + 3 < 14$, cost = 13

(F, 13)

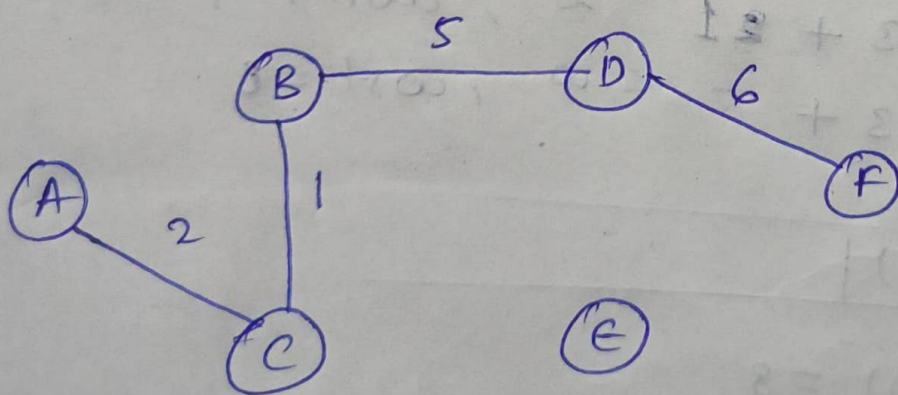
✓, at F :- $d(F) = 13$

$F \rightarrow D : 13 + 6 < 8$, don't update

$F \rightarrow E : 13 + 3 < 14$, don't update

_____ queue empty

Find the shortest path from A to F



$F \rightarrow D \rightarrow B \rightarrow C \rightarrow A$

Q.

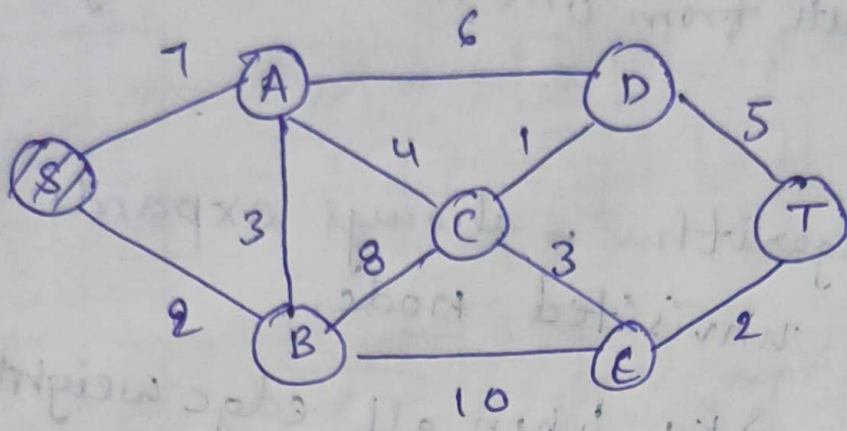


Table :-

Node	cost	previous
S	0	S
A	7 5	B
B	3 2	S
C	8 9	B A
D	6 10	A C
E	10 12	B C
E	0	
T	15 14	E

Formula :- if $d(u) + c(u, v) < d(v)$

then update

(Relaxation formula)

priority queue -

(S, 0)	(A, 7)	(B, 2)
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↓ 1 ↓ 2

1) at S :- $d(S) = 0$

S $\rightarrow$ A :- $0 + 7 < \infty$ ✓, cost = 7

S $\rightarrow$ B :- $0 + 2 < \infty$ ✓, cost = 2

2) at B :- $d(B) = 2$

B $\rightarrow$ S :- $2 + 2 < 0$ ✗, don't update

B $\rightarrow$ A :-

$2 + 3 < 7$ ✓, update cost = 5

B $\rightarrow$ C :-

$2 + 8 < \infty$ ✓, cost = 10

B $\rightarrow$ E :-

$2 + 10 < \infty$ ✓, cost = 12

priority queue -

(A, 5)	(C, 10)	(E, 12)
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↓

3) at A :- $d(A) = 5$

A $\rightarrow$ S :-

$5 + 7 < 0$, don't update
X

A $\rightarrow$ B :-

$5 + 3 < 2$, don't update
X

A $\rightarrow$ D :- $5 + 6 < \infty$, cost = 11

A $\rightarrow$ C :- $5 + 4 < 10$, cost = 9
✓

priority queue

(C, 9)	(E, 12)	(D, 11)
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4) at C :- $d(C) = 9$

C $\rightarrow$ A :- $9 + 4 < 5$ X, don't update

C $\rightarrow$ B :- $9 + 8 < 2$ X, don't update

C $\rightarrow$ D :- $9 + 1 < 11$, cost = 10

C $\rightarrow$ E :- $9 + 3 < 12$ X, don't update

(E, 12)	(D, 10)
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5) at D:- $d(D) = 10$

$D \rightarrow A$:- $10 + 6 < 5 \times$, don't update

$D \rightarrow C$:- $10 + 1 < 9 \times$, don't update

$D \rightarrow T$:- $10 + 5 < \infty$, cost = 15

~~$(E, 12)$~~ | $(T, 15)$

6) at E:- $d(E) = 12$

$E \rightarrow C$:- $12 + 3 < 9 \times$, don't update

~~$E \rightarrow B$~~

$E \rightarrow B$:- $12 + 10 < 2 \times$, don't update

$E \rightarrow T$:- $12 + 2 < 15$, cost = 14

~~$(T, 15)$~~

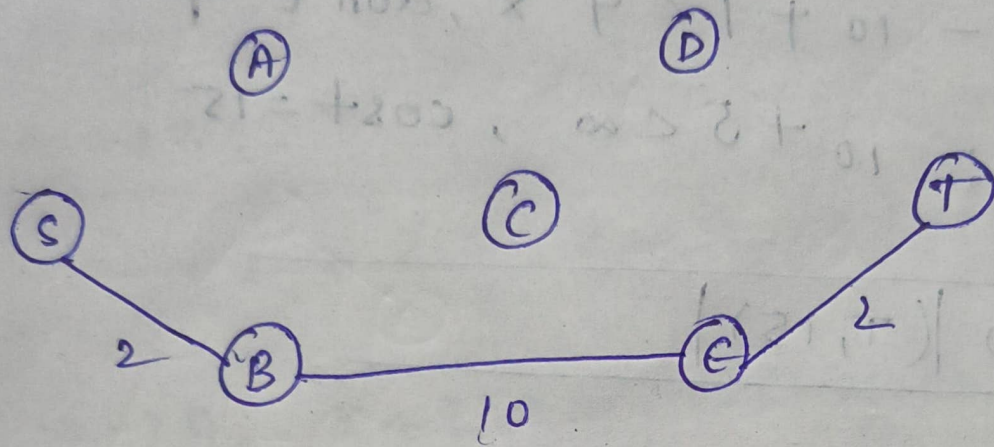
7) at T:- $d(T) = 14$

$T \rightarrow D$:- $14 + 5 < 10 \times$, don't update

$T \rightarrow E$:- $14 + 2 < 12 \times$, don't update

queue empty

Shortest path from $S \rightarrow T$ $\rightarrow$ (a) b $\rightarrow$ c to
 $T \rightarrow e \rightarrow B \rightarrow c$



Shortest path = 14 $\rightarrow$ (a) b $\rightarrow$ c to (c)